fraud detection analysis

October 26, 2026

1 FRAUD TRANSACTIONS ANALYSIS

[2]:
```
# Load the finance transactions fraud dataset
import pandas as pd
data = pd.read_csv(r'C:\Users\HP\Documents\TOP 10␣
↪INDUSTRIES\fraud_detection_project\data\01_finance_transactions_fraud.csv')
```

[3]:
```
# View the data
data.head(8)
```

[3]:

	transaction_id	customer_id	timestamp	amount_kes	merchant_category	channel	country	new_device	is_fraud
0	TX0000001	3154	2025-01-01 00:01:34.000000000	648.08	Utilities	Bank Transfer	KE	0	0
1	TX0000002	3946	2025-01-01 00:55:19.000000000	1332.23	Airtime	M-Pesa	KE	0	0
2	TX0000003	4734	2025-01-01 01:19:24.000000000	329.61	Fuel	ATM	KE	0	0
3	TX0000004	4052	2025-01-01 01:39:02.000000000	4093.17	Restaurants	M-Pesa	KE	0	1
4	TX0000005	2856	2025-01-01 01:41:10.000000000	2564.41	Groceries	Card Online	KE	0	0
5	TX0000006	3612	2025-01-01 01:42:52.000000000	788.43	Utilities	M-Pesa	KE	0	0
6	TX0000007	3191	2025-01-01 02:17:39.000000000	475.42	Utilities	ATM	US	0	0
7	TX0000008	4052	2025-01-01 02:35:27.992152443	6690.51	Electronics	Bank Transfer	KE	0	1

[3]:
```
# data types
data.dtypes
```

[3]:
```
transaction_id          str
customer_id           int64
timestamp               str
amount_kes          float64
merchant_category       str
```

```
channel                  str
country                  str
new_device             int64
is_fraud               int64
dtype: object
```

[5]:
```
# convert timestamp data type to date/time
data["timestamp"] = pd.to_datetime(data["timestamp"],errors="coerce")
```

[7]:
```
# check its changed
data.dtypes
```

[7]:
```
transaction_id                 str
customer_id                  int64
timestamp            datetime64[ns]
amount_kes                 float64
merchant_category              str
channel                        str
country                        str
new_device                   int64
is_fraud                     int64
dtype: object
```

[9]:
```
# put transactions in chronological order
data = data.sort_values("timestamp").reset_index(drop=True)
```

[10]:
```
data.head()
```

[10]:

	transaction_id	customer_id	timestamp	amount_kes	merchant_category	channel	country	new_device	is_fraud
0	TX0000001	3154	2025-01-01 00:01:34	648.08	Utilities	Bank Transfer	KE	0	0
1	TX0000002	3946	2025-01-01 00:55:19	1332.23	Airtime	M-Pesa	KE	0	0
2	TX0000003	4734	2025-01-01 01:19:24	329.61	Fuel	ATM	KE	0	0
3	TX0000004	4052	2025-01-01 01:39:02	4093.17	Restaurants	M-Pesa	KE	0	1
4	TX0000005	2856	2025-01-01 01:41:10	2564.41	Groceries	Card Online	KE	0	0

[11]:
```
# size of the data
data.shape

# 60000 rows and 9 columns
```

[11]:
```
(60000, 9)
```

[12]:
```
# quality of the data
data.isnull().sum()

# there are no missing values
```

[12]:
```
transaction_id       0
customer_id          0
timestamp            0
amount_kes           0
merchant_category    0
channel              0
country              0
new_device           0
is_fraud             0
dtype: int64
```

[13]:
```
data.duplicated().sum()

# there are no duplicates
```

[13]:
```
np.int64(0)
```

[15]:
```
# first transaction timestamp
data['timestamp'].min()
```

[15]:
```
Timestamp('2025-01-01 00:01:34')
```

[16]:
```
# last transaction timestamp
data['timestamp'].max()
```

[16]:
```
Timestamp('2025-12-31 23:47:11')
```

[17]:
```
# are there duplicate transactions?
data['transaction_id'].duplicated().sum()

# there are no duplicate transactions
```

[17]:
```
np.int64(0)
```

[20]:
```
# fraud transactions
data['is_fraud'].sum()

# there are 720 fraud transactions
```

[20]:
```
np.int64(720)
```

[23]:
```
# fraud rate
fraud_rate = ((data['is_fraud'].sum()) / (data['is_fraud'].count())) * 100
```

```
print('fraud rate;',fraud_rate,'%')
```

```
fraud rate; 1.2 %
```

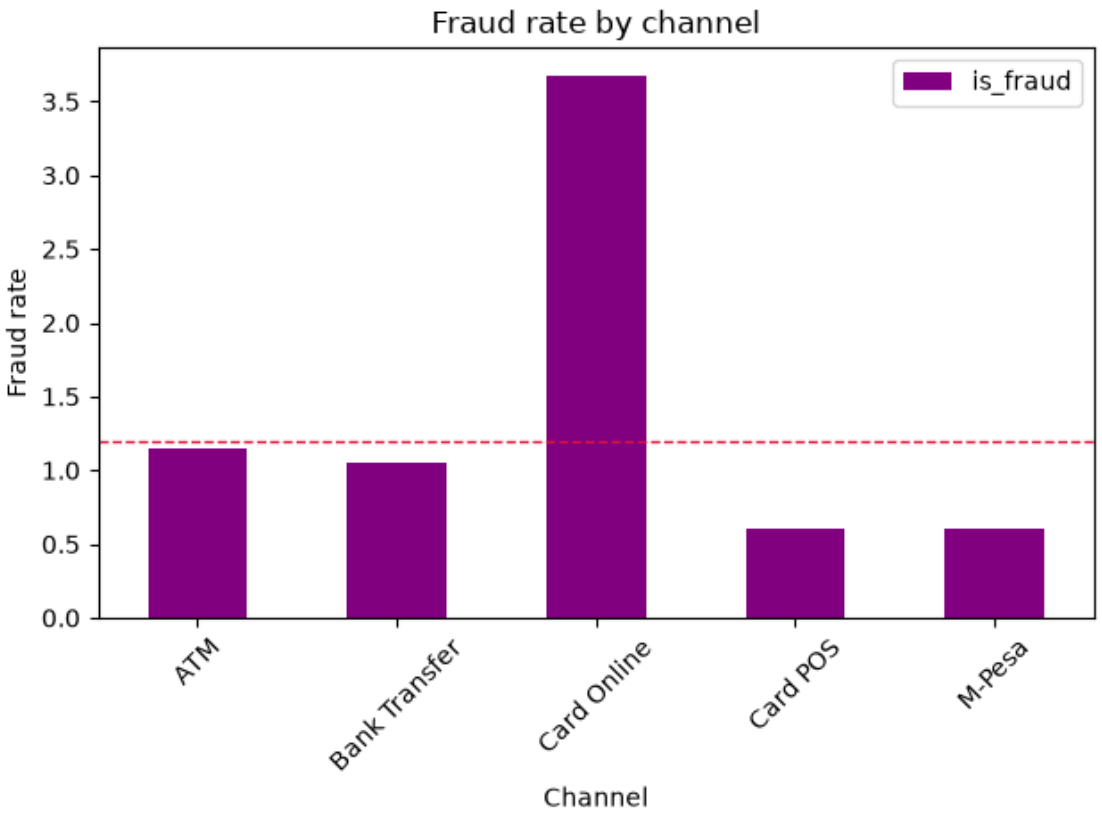
FRAUD RATE BY CHANNEL

[44]:
```
# fraud rate by channel(bar graph)
data['channel'].value_counts()
```

[44]:
```
channel
M-Pesa           23714
Card POS         11924
Card Online       9267
Bank Transfer     9091
ATM               6004
Name: count, dtype: int64
```

[55]:
```
import matplotlib.pyplot as plt
fraud_rate = data['is_fraud'].mean() * 100
channel_fraud_rate = data.groupby('channel')['is_fraud'].mean() * 100
channel_fraud_rate.plot(kind='bar',color='purple')

plt.xlabel('Channel')
plt.ylabel('Fraud rate')
plt.title('Fraud rate by channel')
plt.xticks(rotation=45)
plt.tight_layout()
plt.legend()
plt.axhline(y=fraud_rate, c="crimson", linestyle="--", linewidth=1)
plt.show()
```



[66]:
```
channel_fraud_rate.sort_values(ascending=False)

#Card online has the highest fraud rate of 3.68% way above the average fraud␣
 ↪rate of 1.2%
```

[66]:
```
channel
Card Online      3.679724
ATM              1.149234
Bank Transfer    1.055989
Card POS         0.603824
M-Pesa           0.598802
Name: is_fraud, dtype: float64
```

FRAUD RATE BY HOUR

[56]:
```
# extract hour from the timestamp
data['hour'] = pd.to_datetime(data['timestamp']).dt.hour
data['hour']
```

[56]:
```
0         0
1         0
```

```
2         1
3         1
4         1
         ..
59995    23
59996    23
59997    23
59998    23
59999    23
Name: hour, Length: 60000, dtype: int32
```

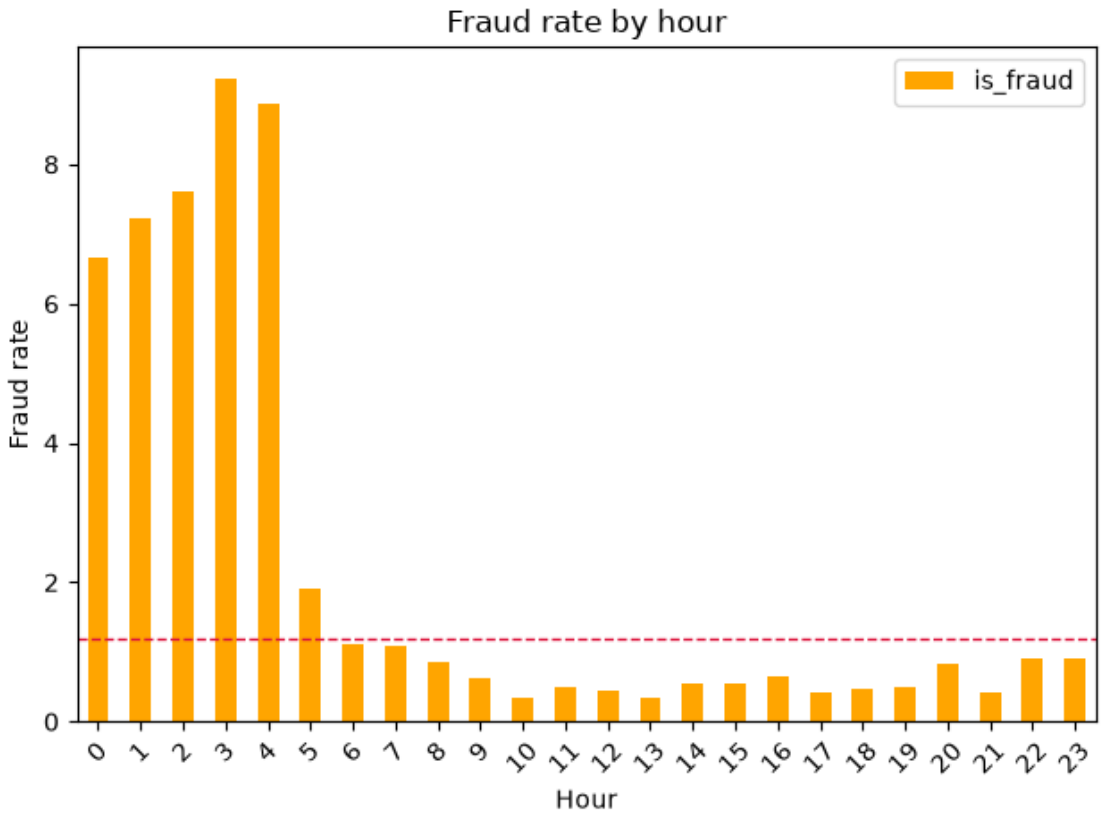
[57]:
```
data.head()
```

[57]:

	transaction_id	customer_id	timestamp	amount_kes	merchant_category	channel	country	new_device	is_fraud	hour
0	TX0000001	3154	2025-01-01 00:01:34	648.08	Utilities	Bank Transfer	KE	0	0	0
1	TX0000002	3946	2025-01-01 00:55:19	1332.23	Airtime	M-Pesa	KE	0	0	0
2	TX0000003	4734	2025-01-01 01:19:24	329.61	Fuel	ATM	KE	0	0	1
3	TX0000004	4052	2025-01-01 01:39:02	4093.17	Restaurants	M-Pesa	KE	0	1	1
4	TX0000005	2856	2025-01-01 01:41:10	2564.41	Groceries	Card Online	KE	0	0	1

[59]:
```
import matplotlib.pyplot as py
fraud_rate = data['is_fraud'].mean() * 100
hour_fraud_rate = data.groupby('hour')['is_fraud'].mean() * 100
hour_fraud_rate.plot(kind='bar',color='orange')

plt.xlabel('Hour')
plt.ylabel('Fraud rate')
plt.title('Fraud rate by hour')
plt.xticks(rotation=45)
plt.tight_layout()
plt.legend()
plt.axhline(y=fraud_rate, c="crimson", linestyle="--", linewidth=1)
plt.show()
```



[65]:
```
# most fraudulent transactions occur in the morning hours from midnight to 5am, all the hours having very high fraud rates avove he average fraud rate
hour_fraud_rate.sort_values(ascending=False)
```

[65]:
```
hour
3     9.222222
4     8.866442
2     7.613636
1     7.223476
0     6.650831
5     1.897733
6     1.123596
7     1.087304
23    0.916870
22    0.895713
8     0.850126
20    0.820966
16    0.659755
9     0.620915
14    0.544523
```

```
15    0.540885
11    0.505849
19    0.505051
18    0.476796
12    0.435053
21    0.422764
17    0.411132
10    0.347991
13    0.340557
Name: is_fraud, dtype: float64
```

2 FRAUD RATE BY COUNTRY

[60]:
```
data.head()
```

[60]:
```
  transaction_id  customer_id           timestamp  amount_kes  \
0      TX0000001         3154 2025-01-01 00:01:34      648.08
1      TX0000002         3946 2025-01-01 00:55:19     1332.23
2      TX0000003         4734 2025-01-01 01:19:24      329.61
3      TX0000004         4052 2025-01-01 01:39:02     4093.17
4      TX0000005         2856 2025-01-01 01:41:10     2564.41

  merchant_category        channel country  new_device  is_fraud  hour
0         Utilities  Bank Transfer      KE           0         0     0
1           Airtime         M-Pesa      KE           0         0     0
2              Fuel            ATM      KE           0         0     1
3       Restaurants         M-Pesa      KE           0         1     1
4         Groceries    Card Online      KE           0         0     1
```

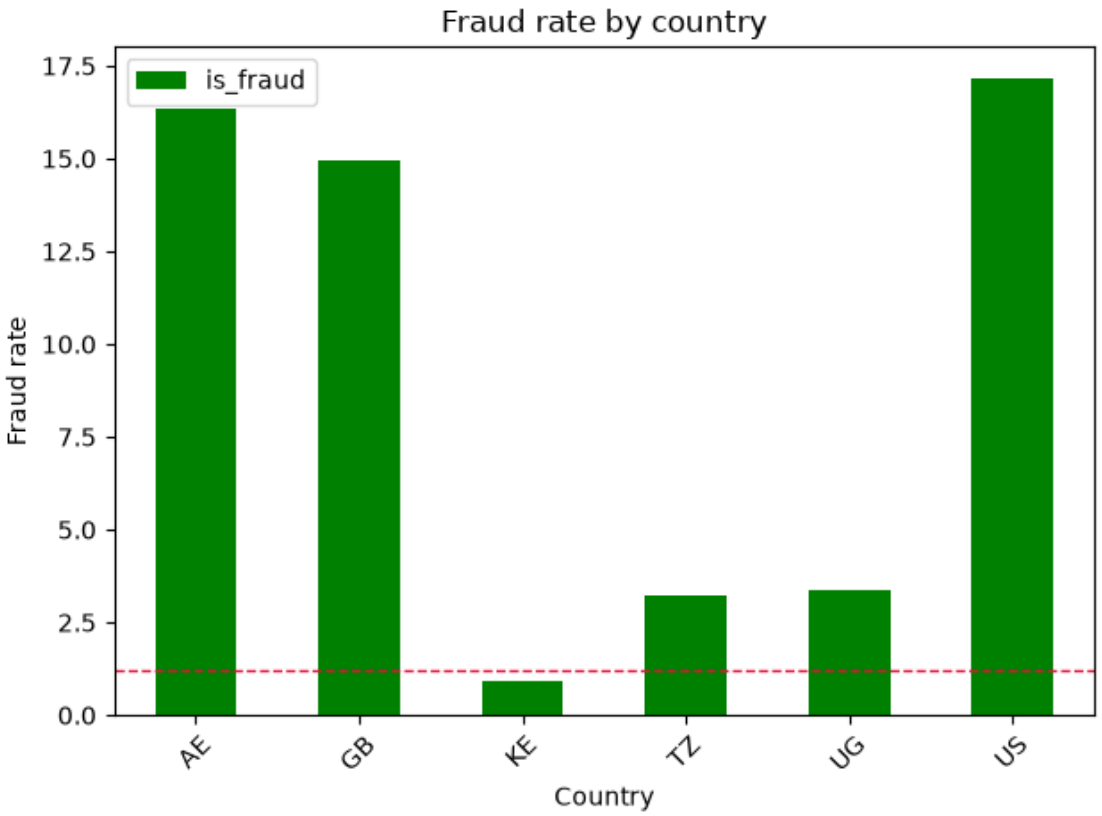
[61]:
```
data['country'].unique()
```

[61]:
```
<ArrowStringArray>
['KE', 'US', 'AE', 'UG', 'TZ', 'GB']
Length: 6, dtype: str
```

[62]:
```
import matplotlib.pyplot as py
fraud_rate = data['is_fraud'].mean() * 100
country_fraud_rate = data.groupby('country')['is_fraud'].mean() *100
country_fraud_rate.plot(kind='bar',color='green')

plt.xlabel('Country')
plt.ylabel('Fraud rate')
plt.title('Fraud rate by country')
plt.xticks(rotation=45)
plt.tight_layout()
plt.legend()
plt.axhline(y=fraud_rate, c="crimson", linestyle="--", linewidth=1)
```

```
plt.show()
```



[64]:
```
# United Arab Emirates, Great Britain, Tanzania, Uganda and the United States␣
 ↪have fraud rates above the average fraud rate of 1.2
country_fraud_rate.sort_values(ascending=False)
# US has the highest fraud rate of 17.15% followed by AE WITH 16.36% AND GB␣
 ↪with 14.97%
# KENYA IS THE ONLY CONTRY WITH A FRAUD RATE OF 0.9%, BELOW THE AVERAGE FRAUD␣
 ↪RATE OF 1.2
```

[64]:
```
country
US    17.153285
AE    16.355140
GB    14.973262
UG     3.352412
TZ     3.243671
KE     0.928956
Name: is_fraud, dtype: float64
```

3 FRAUD RATE BY MERCHANT CATEGORY

[67]:
```
data.head()
```

[67]:
```
  transaction_id  customer_id           timestamp  amount_kes  \
0      TX0000001         3154 2025-01-01 00:01:34      648.08
1      TX0000002         3946 2025-01-01 00:55:19     1332.23
2      TX0000003         4734 2025-01-01 01:19:24      329.61
3      TX0000004         4052 2025-01-01 01:39:02     4093.17
4      TX0000005         2856 2025-01-01 01:41:10     2564.41

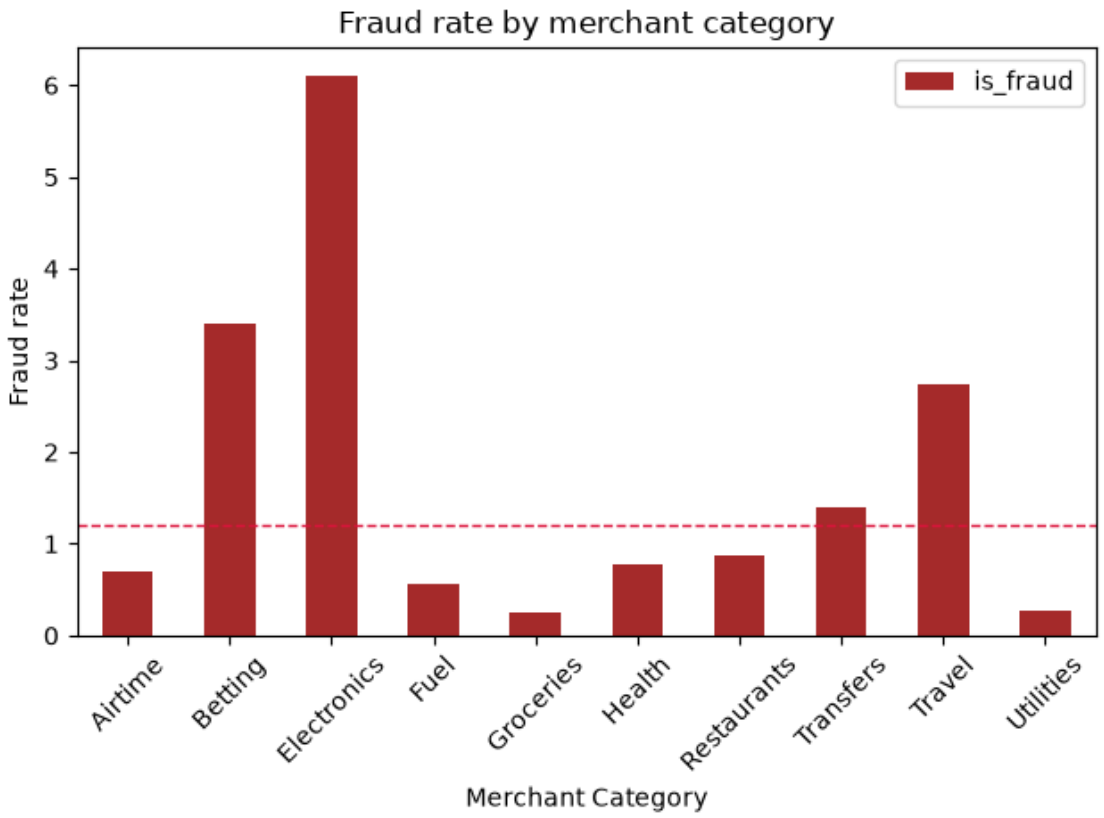
  merchant_category        channel country  new_device  is_fraud  hour
0         Utilities  Bank Transfer      KE           0         0     0
1           Airtime         M-Pesa      KE           0         0     0
2              Fuel            ATM      KE           0         0     1
3       Restaurants         M-Pesa      KE           0         1     1
4         Groceries    Card Online      KE           0         0     1
```

[68]:
```
data['merchant_category'].unique()
```

[68]:
```
<ArrowStringArray>
[  'Utilities',     'Airtime',        'Fuel', 'Restaurants',   'Groceries',
 'Electronics',      'Health',     'Betting',      'Travel',   'Transfers']
Length: 10, dtype: str
```

[69]:
```
import matplotlib.pyplot as py
fraud_rate = data['is_fraud'].mean() *100
merchant_fraud_rate = data.groupby('merchant_category')['is_fraud'].mean()*100
merchant_fraud_rate.plot(kind='bar',color='brown')

plt.xlabel('Merchant Category')
plt.ylabel('Fraud rate')
plt.title('Fraud rate by merchant category')
plt.xticks(rotation=45)
plt.tight_layout()
plt.legend()
plt.axhline(y=fraud_rate, c="crimson", linestyle="--", linewidth=1)
plt.show()
```



[70]:
```
merchant_fraud_rate.sort_values(ascending=False)

# Electronics, Betting, Travel and Transfers are the categories with the most␣
 ↪fraudulent transactions above the average fraud rate of 1.2%
```

[70]:
```
merchant_category
Electronics    6.101915
Betting        3.408360
Travel         2.736156
Transfers      1.390135
Restaurants    0.868839
Health         0.766411
Airtime        0.693745
Fuel           0.567234
Utilities      0.267514
Groceries      0.260045
Name: is_fraud, dtype: float64
```

4 FRAUDULENT TRANSACTIONS AMOUNT

[6]:
```
fraudulent_transactions_amount = data.loc[data['is_fraud'] == 1, 'amount_kes'].
↪sum()

print('Fraudulent Transactions Amount; Ksh.', fraudulent_transactions_amount)
```

```
Fraudulent Transactions Amount; Ksh. 2665780.0
```

[ ]:
```

```